

44.4 Spatial effects

Many regression data sets carry a location: the district a flat is in, the county a case was reported from, the coordinates of a soil sample. Location is rarely of interest in itself; it stands in for everything unmeasured that varies smoothly over space — climate, soil, infrastructure, local labour markets — and leaving it out puts all of that into the error and biases the coefficient of any spatially patterned covariate. Putting it in needs a design block and a penalty, like any smooth covariate. The only question is what “smooth over space” should mean, and there are two answers, one per kind of location data.

44.4.1 The geoadditive model

Definition 44.4.1. Geoadditive model

Let s_i be the location of observation i . The **geoadditive model** is the additive model Eq. 44.1.1 with one extra term,

$$y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \sum_{j=1}^q f_j(z_{ij}) + f_{\text{geo}}(s_i) + \varepsilon_i, \quad \sum_i f_{\text{geo}}(s_i) = 0. \quad (44.4.1)$$

Two cases are distinguished by the kind of location.

- (a) (*Areal data.*) $s_i \in \{1, \dots, D\}$ names one of D regions. The design is the **incidence matrix** $\mathbf{Z}_{\text{geo}}$ with $(\mathbf{Z}_{\text{geo}})_{is} = 1$ if $s_i = s$ and 0 otherwise, the coefficient γ_s is the effect of region s , and the penalty is built from a **neighbourhood structure**: a symmetric relation $s \sim t$ saying which regions share a border.
- (b) (*Point-referenced data.*) $s_i \in \mathbb{R}^2$ is a coordinate pair. The design is $(\mathbf{Z}_{\text{geo}})_{il} = r(\|s_i - \kappa_l\|)$ for a set of **knots** $\kappa_1, \dots, \kappa_L$ and a radial function r , and the penalty is $\mathbf{P}_{\text{geo}} = \{r(\|\kappa_l - \kappa_m\|)\}_{lm}$.

In (a) the map is a graph and smoothness means that neighbouring regions do not differ by much; in (b) the map is a plane and smoothness means what it means for a surface. Both are pairs $(\mathbf{Z}_{\text{geo}}, \mathbf{P}_{\text{geo}})$, so [Sections 44.1](#) and [44.2](#) apply without a word of change.

44.4.2 Markov random fields on a map

For areal data, write n_s for the number of neighbours of region s and define the **neighbourhood penalty**

$$\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} = \sum_{s \sim t} (\gamma_s - \gamma_t)^2, \quad K_{st} = \begin{cases} n_s, & t = s, \\ -1, & t \sim s, \\ 0, & \text{otherwise,} \end{cases} \quad (44.4.2)$$

where the sum runs over unordered neighbour pairs: the first-difference penalty of Definition 43.3.1 with “next knot” replaced by “neighbouring region”, the two coinciding exactly on a chain $1 \sim 2 \sim \dots \sim D$.

Example. Four regions in a row

Let four districts lie in a row, $1 \sim 2 \sim 3 \sim 4$:

$$\mathbf{P} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix},$$

with $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} = (\gamma_1 - \gamma_2)^2 + (\gamma_2 - \gamma_3)^2 + (\gamma_3 - \gamma_4)^2$, rank 3 and null space $\text{span}\{\mathbf{1}\}$. If district 4 is instead an island with no neighbours, its row and column vanish, the rank drops to 2 and the null space is spanned by $(1, 1, 1, 0)^\top$ and $(0, 0, 0, 1)^\top$: the island is not penalized at all and is estimated from its own data alone.

Proposition 44.4.1. The neighbourhood penalty

Let the regions $1, \dots, D$ carry a symmetric neighbourhood relation with no region its own neighbour, and let $\mathbf{P}$ be as in Eq. 44.4.2.

- (a) $\mathbf{P}$ is symmetric and nonnegative definite, and $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} = \sum_{s \sim t} (\gamma_s - \gamma_t)^2$.
- (b) $\mathcal{N}(\mathbf{P})$ is spanned by the indicator vectors of the connected components of the neighbourhood graph. In particular $\text{rank}(\mathbf{P}) = D - c$ where c is the number of components, and $\mathcal{N}(\mathbf{P}) = \text{span}\{\mathbf{1}\}$ iff the graph is connected.
- (c) Treating $\exp\{-\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} / (2\tau^2)\}$ as a density in $\boldsymbol{\gamma}$ on $\mathcal{N}(\mathbf{P})^\perp$, the conditional distribution of γ_s given the others, *for a region with $n_s \geq 1$* , is normal with mean $n_s^{-1} \sum_{t \sim s} \gamma_t$, the average of its neighbours, and variance τ^2 / n_s . The prior is therefore a **Gaussian Markov random field**: each region is shrunk towards its neighbours, and a region with many neighbours is shrunk harder. A region with $n_s = 0$ contributes nothing to the quadratic form: its coefficient is unpenalized, its conditional is flat, and it is estimated from its own data alone (Example (Four regions in a row), Exercise (A1)).

Proof.

- (a) Symmetry is immediate. Each unordered neighbour pair is counted twice in the double sum over ordered pairs, so

$$\sum_{s \sim t} (\gamma_s - \gamma_t)^2 = \frac{1}{2} \sum_s \sum_{t \sim s} (\gamma_s^2 - 2\gamma_s \gamma_t + \gamma_t^2).$$

The first inner sum gives $\sum_s n_s \gamma_s^2$, and so does the third, because $\sum_s \sum_{t \sim s} \gamma_t^2 = \sum_t n_t \gamma_t^2$ by symmetry of the relation. Hence the right-hand side is $\sum_s n_s \gamma_s^2 - \sum_s \sum_{t \sim s} \gamma_s \gamma_t = \boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma}$. Nonnegativity follows because the left-hand side is a sum of squares.

- (b) By (a), $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} = 0$ iff $\gamma_s = \gamma_t$ across every edge, iff $\boldsymbol{\gamma}$ is constant on each connected component. Since $\mathbf{P}$ is nonnegative definite, that is equivalent to $\mathbf{P} \boldsymbol{\gamma} = \mathbf{0}$ by Theorem 1.7.4, so the null space is the span of the component indicators, which are independent because the components are disjoint.
- (c) If $n_s = 0$ then row s of $\mathbf{P}$ is zero and γ_s does not appear in the quadratic form, which

is the last sentence. For $n_s \geq 1$, collect the terms of $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma}$ that involve γ_s :

$$n_s \gamma_s^2 - 2\gamma_s \sum_{t \sim s} \gamma_t + \text{const} = n_s \left(\gamma_s - \frac{1}{n_s} \sum_{t \sim s} \gamma_t \right)^2 + \text{const}'.$$

Dividing by $2\tau^2$ and exponentiating gives the normal density with the stated mean and variance τ^2/n_s . ■

Part (c) is the reason for the name. The penalty does not shrink a region towards zero — that would be the ridge penalty $\mathbf{P} = \mathbf{I}$ — but towards its neighbourhood, so a district with few observations borrows strength from those around it while the overall level of the map stays free. That level is the one dimension the centring constraint of Proposition 44.1.1(c) removes on a connected map; it belongs to the intercept.

Remark (Variants). Three variants fit the same framework. Weighting each pair by w_{st} (border length, inverse centroid distance) replaces n_s by $\sum_{t \sim s} w_{st}$ and -1 by $-w_{st}$. Adding $\epsilon \mathbf{I}$ makes the penalty positive definite and the prior proper, at the price of shrinking the map towards zero as well as towards itself. And a *second*, unstructured term with $\mathbf{P} = \mathbf{I}$ on the same regions gives the convolution model of Besag, York and Mollié (1991), separating smooth spatial variation from region-specific noise.

44.4.3 Kriging bases for coordinates

For points rather than regions, smoothness comes from the theory of stationary random fields, which is to say from [Part VII](#). Let the spatial effect be a realization of a mean-zero process $\{U(s)\}$ with $\text{Cov}\{U(s), U(t)\} = \tau^2 r(\|s - t\|)$: a covariance model in the sense of Definition 33.4.1, on a continuous index set.

Proposition 44.4.2. Low-rank kriging is a penalized term

Let $\mathbf{R} = \{r(\|\kappa_l - \kappa_m\|)\}_{lm}$ be the correlation matrix of the field at L knots, assumed positive definite, and let $\mathbf{u} = \{U(\kappa_1), \dots, U(\kappa_L)\}^\top \sim N_L(\mathbf{0}, \tau^2 \mathbf{R})$.

- (a) The best linear predictor of $U(s)$ from $\mathbf{u}$ is $\hat{U}(s) = \mathbf{r}(s)^\top \mathbf{R}^{-1} \mathbf{u}$ with $\mathbf{r}(s) = \{r(\|s - \kappa_1\|), \dots, r(\|s - \kappa_L\|)\}^\top$. Writing $\boldsymbol{\gamma} = \mathbf{R}^{-1} \mathbf{u}$, the fitted spatial effect at the observations is $\mathbf{Z}_{\text{geo}} \boldsymbol{\gamma}$ with $(\mathbf{Z}_{\text{geo}})_{il} = r(\|s_i - \kappa_l\|)$, and $\boldsymbol{\gamma} \sim N_L(\mathbf{0}, \tau^2 \mathbf{R}^{-1})$.
- (b) Consequently the log-density of $\boldsymbol{\gamma}$ contributes $\boldsymbol{\gamma}^\top \mathbf{R} \boldsymbol{\gamma} / (2\tau^2)$, so the penalized criterion Eq. 44.1.4 with $\mathbf{P}_{\text{geo}} = \mathbf{R}$ and $\lambda_{\text{geo}} = \sigma^2 / \tau^2$ has as its minimizer the posterior mode of $\boldsymbol{\gamma}$ — the kriging predictor restricted to the span of the L knots.
- (c) Taking every distinct location as a knot recovers exact kriging, that is, generalized least squares with the covariance $\tau^2 \mathbf{R} + \sigma^2 \mathbf{I}$ of [Chapter 31](#); taking $L \ll n$ knots is the low-rank approximation that makes the term cheap.

Proof.

- (a) By Theorem 14.2.8 (or Theorem 32.3.1 with $\mathbf{X}$ absent) the best linear predictor is $\text{Cov}\{U(s), \mathbf{u}\} \text{Cov}(\mathbf{u})^{-1} \mathbf{u} = \tau^2 \mathbf{r}(s)^\top (\tau^2 \mathbf{R})^{-1} \mathbf{u}$, the stated formula; evaluating at $s = s_i$

stacks into $\mathbf{Z}_{\text{geo}}\mathbf{R}^{-1}\mathbf{u}$, and the distribution of $\boldsymbol{\gamma} = \mathbf{R}^{-1}\mathbf{u}$ follows from Theorem 2.2.1.

- (b) $-\log$ density of $N_L(\mathbf{0}, \tau^2\mathbf{R}^{-1})$ is $\boldsymbol{\gamma}^\top \mathbf{R} \boldsymbol{\gamma} / (2\tau^2)$ up to a constant. Adding it to $\|\mathbf{y} - \dots\|^2 / (2\sigma^2)$ and multiplying by $2\sigma^2$ gives Eq. 44.1.4 with the stated λ .
- (c) With $L = n$ distinct locations, $\mathbf{Z}_{\text{geo}} = \mathbf{R}$ and the term contributes $\mathbf{R}\boldsymbol{\gamma} = \mathbf{u}$ with $\mathbf{u}$ penalized by $\mathbf{u}^\top \mathbf{R}^{-1}\mathbf{u}$; minimizing over $\mathbf{u}$ is the mixed model of Definition 32.2.1 with $\mathbf{Z} = \mathbf{I}$, $\mathbf{G} = \tau^2\mathbf{R}$ and $\mathbf{R} = \sigma^2\mathbf{I}$, whose marginal covariance is $\tau^2\mathbf{R} + \sigma^2\mathbf{I}$ (Eq. 32.2.2).

■

Idea. A spatial term is a random effect with a structured covariance — not an analogy: by Proposition 44.4.2 the penalty is the inverse prior covariance, and Section 44.5 makes that general. The Markov random field Eq. 44.4.2 says the same with a sparse precision matrix in place of a dense covariance, which is why it is cheap on large maps.

If $\mathbf{R}$ and $\mathbf{Z}_{\text{geo}}$ are to be sparse, the correlation function must have compact support or decay fast. The example below uses the Wendland function $r(h) = (1 - h/\rho)_+^4(1 + 4h/\rho)$, positive definite in two dimensions and zero beyond the range ρ .

44.4.4 A map

Example. Smoothing a district map

A simulated country is a 12×12 lattice of 144 districts with between 1 and 8 observations each, 604 in all. The response has mean $2 + 0.7x + \text{surface}(s)$, with x an ordinary covariate and the surface a smooth function of the district centroid that the model does not know; the errors are normal with standard deviation 0.9.

Figure 44.4.1 compares three fits. Unpenalized district effects use 143.0 degrees of freedom and give the noisy map of panel (b), with root mean squared error 0.579 against the true surface. The neighbourhood penalty Eq. 44.4.2, with λ from the restricted likelihood, uses 45.5 and brings that down to 0.170. A kriging term on 21 knots uses 17.7 and gives 0.228: smoother still, and slightly too smooth, because the true surface has features the coarse knot grid cannot represent. The coefficient of x comes out at 0.581 with a standard error of 0.042 — against a true 0.7, about 2.8 standard errors low on this draw — and the residual standard deviation at 0.953.

The map and its penalty are built from the neighbourhood list alone:

```
import numpy as np

def lattice(side):
    """Rook adjacency on a side x side grid of districts, and the district centroids
    ."""
    coords = np.array([[c + 0.5, r + 0.5] for r in range(side) for c in range(side)
    ])
    neighbours = []
    for s in range(side * side):
        r, c = divmod(s, side)
        nb = [(r + dr) * side + (c + dc) for dr, dc in ((-1, 0), (1, 0), (0, -1),
        (0, 1))]
```

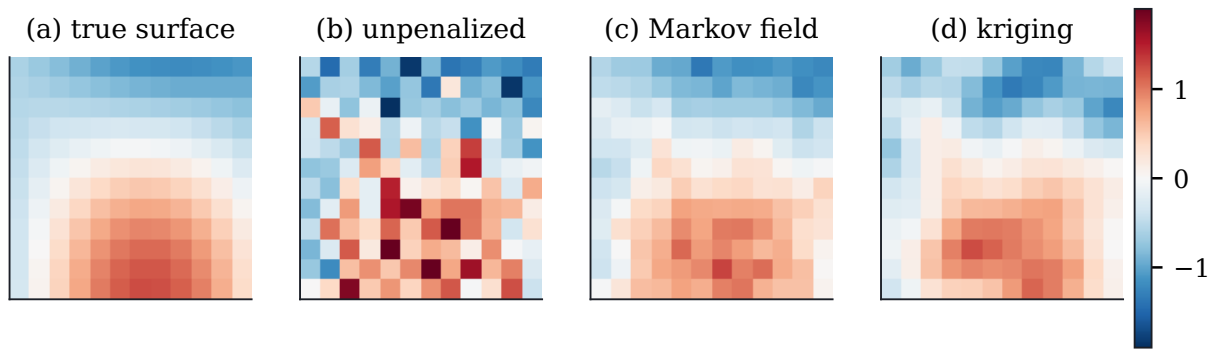


Figure 44.4.1: A simulated district map, all panels on one colour scale. (a) the surface used to generate the data. (b) unpenalized district effects. (c) the Markov random field Eq. 44.4.2. (d) a kriging term on 21 knots.

```

        if 0 <= r + dr < side and 0 <= c + dc < side]
        neighbours.append(sorted(nb))
    return coords, neighbours

def neighbourhood_penalty(neighbours):
    """K = D - W: the number of neighbours on the diagonal, -1 for each neighbour
    pair."""
    d = len(neighbours)
    K = np.zeros((d, d))
    for s, nbrs in enumerate(neighbours):
        K[s, s] = len(nbrs)
        for t in nbrs:
            K[s, t] = -1.0
    return K

```

[Run this code online](#)

Warning (Spatial confounding). A spatial term competes with every covariate that is itself spatially patterned: they are concure in the sense of Definition 44.2.1. In [Example \(Smoothing a district map\)](#) the covariate x was generated independently of location, so nothing was lost. When a covariate varies smoothly over the map — income across districts, exposure across counties — the spatial term can absorb a large part of its effect, and the fitted coefficient moves, sometimes dramatically, when the term is added. Neither fit is “the” answer: without the spatial term all spatial variation is attributed to the covariate, with it the smooth part goes to location. Which is right is a question about unmeasured confounders, and the fit has no opinion; [Chapter 25](#) has the vocabulary. Report both, and say which unmeasured causes the spatial term stands in for.

44.4.5 Exercises

44.4.5.1 A. Check your understanding

Exercise. A1

A map has two islands, each a connected group of districts, with no neighbour pairs between them. What is $\text{rank}(\mathbf{P})$, what does the penalty leave unidentified, and what does the centring

constraint of Proposition 44.1.1(c) fix?

44.4.5.2 B. Practice

Exercise. B1

Show that on a chain $1 \sim 2 \sim \dots \sim D$, the penalty Eq. 44.4.2 equals the first-difference penalty $\boldsymbol{\gamma}^\top \mathbf{D}_1^\top \mathbf{D}_1 \boldsymbol{\gamma}$ of Definition 43.3.1, and that the second-difference penalty is *not* of the form Eq. 44.4.2 for any neighbourhood structure with nonnegative off-diagonal weights.

Exercise. B2

Using Proposition 44.4.1(c), show that if district s has n_s neighbours and its own m_s observations with mean $\bar{y}_s$, and all other effects are held fixed, the penalized criterion is minimized over γ_s at a weighted average of $\bar{y}_s$ and the neighbour mean, with weights m_s and λn_s . What happens when $m_s = 0$?

Solution. Up to constants the terms of Eq. 44.1.4 involving γ_s are $m_s(\bar{y}_s - \gamma_s)^2 + \lambda n_s(\gamma_s - \bar{\gamma}_{(s)})^2$ with $\bar{\gamma}_{(s)} = n_s^{-1} \sum_{t \sim s} \gamma_t$, so $\hat{\gamma}_s = (m_s \bar{y}_s + \lambda n_s \bar{\gamma}_{(s)}) / (m_s + \lambda n_s)$. With $m_s = 0$ the district is set to its neighbour average: the map interpolates over the gap.

44.4.5.3 C. Going deeper

Exercise. C1

Let $\mathbf{v}$ be the vector of a covariate that is constant within each district, so that $\mathbf{v} \in \mathcal{C}(\mathbf{Z}_{\text{geo}})$ for the incidence design. Show that in the *unpenalized* geoadditive model the coefficient of $\mathbf{v}$ is not identified at all, and that with a penalty it is identified but its estimate depends on λ_{geo} , tending to the no-spatial-term estimate as $\lambda_{\text{geo}} \rightarrow \infty$. What does this say about reporting such a coefficient?